

Udit Rawat

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SUMMARY

AI Engineer with production experience building RAG pipelines, fine-tuning and deploying LLMs (LoRA/QLoRA) on edge hardware, and shipping multi-agent systems at scale. Springer-published on LLM adaptation; promoted from intern to engineer in under 2 months.

TECHNICAL SKILLS

Languages	Python (async), TypeScript, SQL, Shell
AI & ML	PyTorch, HF Transformers, LangChain, LangGraph, DSPy, RAG (pgvector, Qdrant), LLM Fine-tuning (LoRA/QLoRA), RAGAS
Model Optimization	GGUF, llama.cpp, ONNX, INT8/NF4 Quantization, Knowledge Distillation
Cloud & Backend	GCP, Azure, AWS, Cloudflare Workers/Pages, FastAPI, Docker, Kubernetes
MLOps & Data	LiteLLM, Langfuse, ClickHouse, PostgreSQL, CI/CD

EXPERIENCE

AI Engineer, TapHealth	New Delhi, India	Apr 2025 – Present
- Built and shipped an offline diabetes-guidance SLM running fully on-device on Snapdragon 680 hardware, in partnership with Eli Lilly: 3× smaller at 380 MB, under 890 ms latency at 25 tok/s, 98% accuracy on out-of-distribution refusal. Fine-tuned Qwen 0.6B/1.5B with LoRA, quantized to GGUF Q4_K_M, and built a separate ONNX INT8 MPNet classifier for OOD detection. - Built a stateful 5-agent coaching system with LangGraph and Mem0/Hindsight memory across 25 tools, served via FastAPI with hybrid retrieval over pgvector/Qdrant. Built an async OCR/PDF pipeline streamed over SSE WebSockets, cutting report processing from 40s to 15s. - Cut LLM costs 54% by designing a centralized LiteLLM gateway routing requests across OpenAI, Anthropic, and open-source models for 65+ agentic workflows spanning 5+ microservices, with structured outputs via Instructor and full request tracing via Langfuse. - Rebuilt a monolithic 812-line rules-based nudge pipeline into a decoupled DSPy-based GenAI architecture (habit classifier separated from content generator), plus a standalone summary-generation service, serving 5K+ paid and free users. Built a week-0/week-N onboarding planner and daily scheduler for weekly planning, and a ClickHouse-backed analytics endpoint aggregating glucose, HbA1c, and meal data, with full CI/CD.		
AI Intern, TapHealth	New Delhi, India	Feb 2025 – Mar 2025
Built ARIMA/SARIMA/LSTM forecasting models for patient engagement and generated synthetic clinical data using NHANES + Q-learning, with LIME-based explainability. <i>Promoted to AI Engineer after 2 months.</i>		
Project Intern, Perception & Intelligence Lab, IIT Kanpur		Jul 2024 – Oct 2024
Curated a 200K+ image dataset and optimized diffusion-based inpainting models, achieving 11.72 dB PSNR and 0.41 SSIM.		

PROJECTS

Hierarchical Reasoner (<i>Python, PyTorch</i>) github.com/udit-rawat/hierarchical-reasoner From-scratch PyTorch reproduction of the Hierarchical Reasoning Model (arXiv:2506.21734) with dual-module GRU and ACT Q-learning adaptive halting. Reached 100% maze-path accuracy by epoch 20 with 66.7% compute reduction versus baseline; validated across 8 controlled experiments with 49 pytest cases.
Animatic (<i>Python, Claude/Gemini API</i>) github.com/udit-rawat/animatic Multi-stage AI video pipeline (LLM scripting, Gemini TTS narration, word-level alignment, HyperFrames rendering) with quota-aware multi-provider routing across Claude, Gemini, and OpenRouter. Automated daily publishing generated 66K+ organic views and 590 new followers in the last 40 days alone, with zero paid promotion.

PUBLICATIONS

Efficient Adaptation of Lightweight LLMs with T-Free Tokenization Boosting Algorithms for Class Imbalance	ICAAI 2025, Springer SSWC 2025
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EDUCATION

B.Tech in AI & ML, VIPS-TC, GGSIPU, New Delhi	2021 – 2025	GPA: 9.1/10
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